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Notebook 03 — Feature Engineering & Engagement Scoring

Project: Ad Engagement Analysis

Goal: Build a custom business metric to rank ad effectiveness

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In this notebook, we move beyond just counting interactions. We are going to build a custom **Engagement Effectiveness Score**.

The Business Logic:

Not all interactions are equal.

- Base Comment = **1 point** (Minimum effort)
- Emoji Used = **+2 points** (Shows active emotional reaction)
- Hashtags Used = **+1 point per hashtag** (Shows intent to amplify/share the ad visually)

Combining these gives us a weighted score, allowing us to find the campaigns that truly performed the best!

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import warnings
```

```
warnings.filterwarnings('ignore')
sns.set_theme(style="whitegrid")

# Load Dataset
try:
    df = pd.read_csv('../data/comments_cleaned.csv')
    print("Dataset 'comments_cleaned.csv' loaded successfully.")
except FileNotFoundError:
    print("Dataset not found error.")

# Prepare target columns if they exist
if 'emoji used' in df.columns:
    df['emoji_binary'] = df['emoji used'].apply(lambda x: 1 if str(x).strip().lower() == 'yes' else 0)
else:
    df['emoji_binary'] = 0

if 'Hashtags used count' in df.columns:
    df['hashtags_cnt'] = df['Hashtags used count'].fillna(0).astype(int)
else:
    df['hashtags_cnt'] = 0

display(df[['comment', 'emoji used', 'Hashtags used count']].head())
```

Dataset 'comments_cleaned.csv' loaded successfully.

	comment	emoji used	Hashtags used count
0	unde at dolore	yes	1
1	quae ea ducimus	no	2
2	alias a voluptatum	no	4
3	facere suscipit sunt	yes	2
4	totam eligendi quaerat	yes	1

Constructing the Engagement Effectiveness Score

Let's encode the business logic directly into mathematics.

```
In [2]: # The Business Logic Math
# Base Score (existence of comment) = 1
base_score = 1

# Emoji Used = +2 points
emoji_score = df['emoji_binary'] * 2

# Hashtag Used = +1 point per hashtag
hash_score = df['hashtags_cnt'] * 1

df['Engagement_Effectiveness_Score'] = base_score + emoji_score + hash_score

# Check how it looks
display(df[['comment', 'emoji used', 'Hashtags used count', 'Engagement_Effectiveness_Score']].head(10))
```

	comment	emoji used	Hashtags used count	Engagement_Effectiveness_Score
0	unde at dolorem	yes	1	4
1	quae ea ducimus	no	2	3
2	alias a voluptatum	no	4	5
3	facere suscipit sunt	yes	2	5
4	totam eligendi quaerat	yes	1	4
5	vitae quia aliquam	no	2	3
6	exercitationem occaecati neque	yes	0	3
7	sint ad fugiat	no	5	6
8	nesciunt aut nesciunt	no	6	7
9	laudantium ut nostrum	yes	2	5

Ad Campaign Ranking

Let's group by `Photo id` (which acts as our ad campaign identifier) and sum the Engagement Effectiveness Scores to find our top-performing ads.

```
In [3]: # Grouping Campaigns
if 'Photo id' in df.columns:
    campaign_scores = df.groupby('Photo id')['Engagement_Effectiveness_Score'].sum().reset_index()
    campaign_scores = campaign_scores.rename(columns={'Engagement_Effectiveness_Score': 'Total Ad Score'})

    # Sort campaigns by score descending
    campaign_scores = campaign_scores.sort_values(by='Total Ad Score', ascending=False)

    print("Top 10 Most Effective Ad Campaigns:")
    display(campaign_scores.head(10))

    # Visualization
    plt.figure(figsize=(10,6))
    sns.barplot(
        x='Photo id',
        y='Total Ad Score',
        data=campaign_scores.head(10),
        order=campaign_scores['Photo id'].head(10),
        palette='viridis'
    )
    plt.title('Top 10 Most Effective Ad Campaigns (Based on Custom Engagement Logic)')
    plt.xlabel('Ad Campaign ID (Photo id)')
    plt.ylabel('Total Evaluated Score')
    plt.show()
else:
    print("Cannot rank: 'Photo id' column missing.")
```

Top 10 Most Effective Ad Campaigns:

Photo id	Total Ad Score	
156	157	173
246	247	172
12	13	171
7	8	168
145	146	164
224	225	159
142	143	159
28	29	159
174	175	156
199	200	156

